📊 DATA ANALYTICS PROJECT

Uber Supply–Demand Gap Analysis

An end-to-end data analytics project analyzing ride request data to identify supply-demand gaps, cancellation trends, and peak demand hours.

TOOLS
Excel · MySQL · Python

FOCUS
EDA & Business Insights

DOMAIN
Ride-Hailing Operations

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Project Objective

Understanding the core business problem and analytical goals

The goal of this project was to analyze Uber ride request data to identify **supply-demand gaps**, **ride cancellation trends**, **cab unavailability patterns**, and **peak demand hours**.

🎯 Key Business Questions

- When does Uber face the highest ride demand?
- During which hours are rides cancelled most?
- When are cabs most unavailable?
- Which pickup locations face the highest operational issues?
- How can Uber improve driver allocation and ride completion rates?

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Tools & Technologies

Technology stack used across the analytics pipeline

Excel → MySQL → Python → Visualization → Insights

TOOL	PURPOSE
Excel	Data cleaning and preprocessing
MySQL	SQL analysis and business queries
Python	Exploratory Data Analysis (EDA)
Pandas	Data manipulation and aggregation
Matplotlib	Data visualization
Seaborn	Statistical plotting and charts
Jupyter	EDA workflow and documentation

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Dataset Overview

Understanding the structure and scope of the data

8	2	3
DATA FIELDS	PICKUP ZONES	RIDE STATUSES

📁 Dataset Fields

- Request ID
- Pickup Point
- Driver ID
- Ride Status
- Request Timestamp
- Drop Timestamp
- Hour
- Time Slot

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Data Cleaning in Excel

Preprocessing steps performed before analysis

🧹 Cleaning Steps

- Checked for missing values
- Verified duplicate records
- Removed inconsistencies
- Handled NA values in Driver ID
- Handled NA values in Drop Timestamp
- Verified date/time formatting

➕ Derived Columns

- **Hour** — Extracted from Request Timestamp
- **Time Slot** — Categorized into Morning, Afternoon, Evening, Night

✅ Data quality was verified as suitable for SQL and Python analysis after preprocessing.

🔍 Key Observations

- Missing Driver IDs were linked to **"No Cars Available"** ride status
- Drop Timestamp was unavailable for **cancelled rides** and **failed requests**

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SQL Analysis in MySQL

Business queries executed on the cleaned dataset

🗄️ SQL Tasks Performed

- Created database and tables
- Imported cleaned CSV dataset
- Executed business analysis queries
- Performed aggregation and grouping analysis

📝 Key Queries Executed

- Total ride requests
- Ride status distribution
- Requests by pickup point
- Requests by time slot
- Peak demand hour analysis
- Cancellation trend analysis
- "No Cars Available" analysis
- Pickup point vs ride status
- Supply-demand gap analysis

💡 SQL Insights

- Evening hours showed the highest number of **"No Cars Available"** cases
- Morning hours had **higher ride cancellation** rates
- Airport requests faced higher **cab unavailability**
- City requests experienced more **cancellations**
- Demand peaked during **office commute timings**

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Exploratory Data Analysis

EDA performed using Python in Jupyter Notebook

📦 Libraries Used

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

🔎 Dataset Inspection

```
df.head() # Preview first 5 rows
df.info() # Column types and non-null counts
df.describe() # Statistical summary
df.isnull().sum() # Missing value count per column
```

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Visualizations & Insights

Interactive charts revealing key patterns in the data

Chart 1 — Ride Status Distribution

Proportion of completed, cancelled, and unavailable rides

💡 **Key Insight:** "No Cars Available" formed a major percentage of failed rides, indicating a significant supply shortage.

Chart 2 — Ride Status by Time of Day

Comparing ride outcomes across different time periods

💡 **Key Insight:** Evening hours showed the highest cab unavailability, while morning hours saw more cancellations.

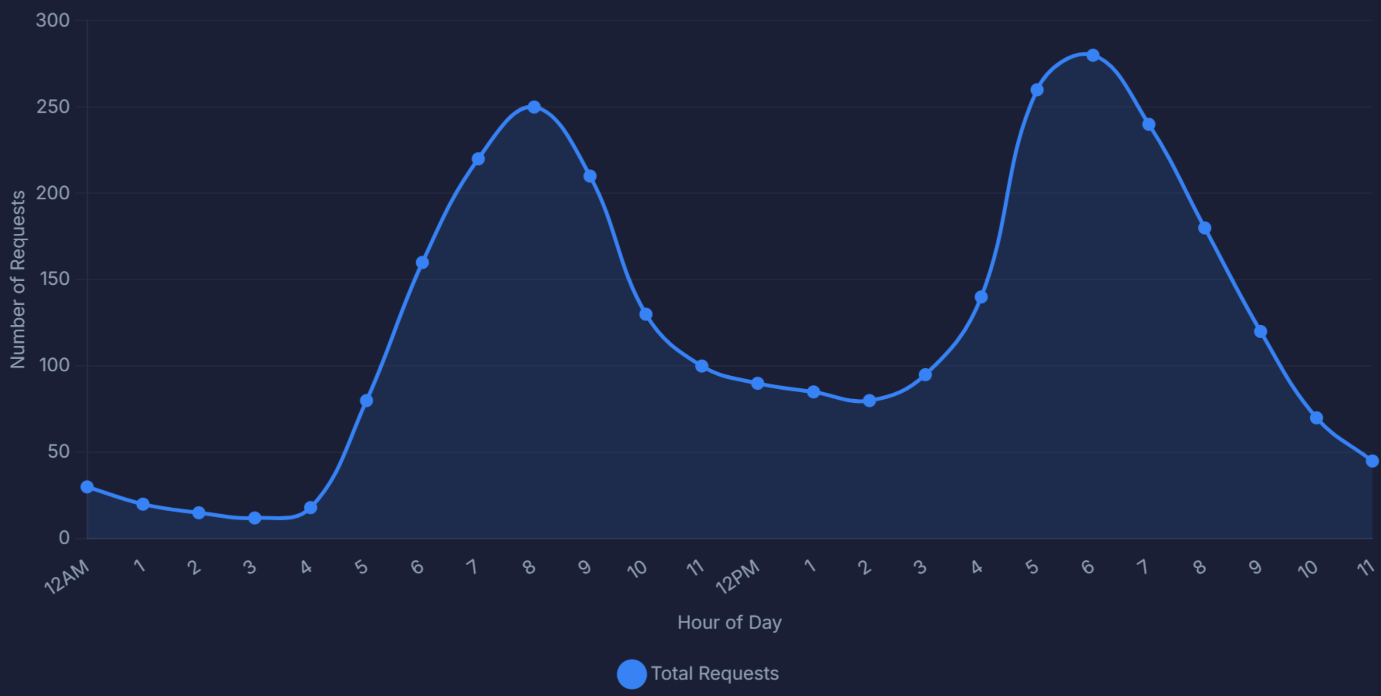
Chart 3 — Ride Status by Pickup Point

Comparing Airport vs City operational performance

💡 **Key Insight:** Airport pickups experienced higher cab unavailability. City pickups experienced more cancellations.

Chart 4 — Ride Requests by Hour

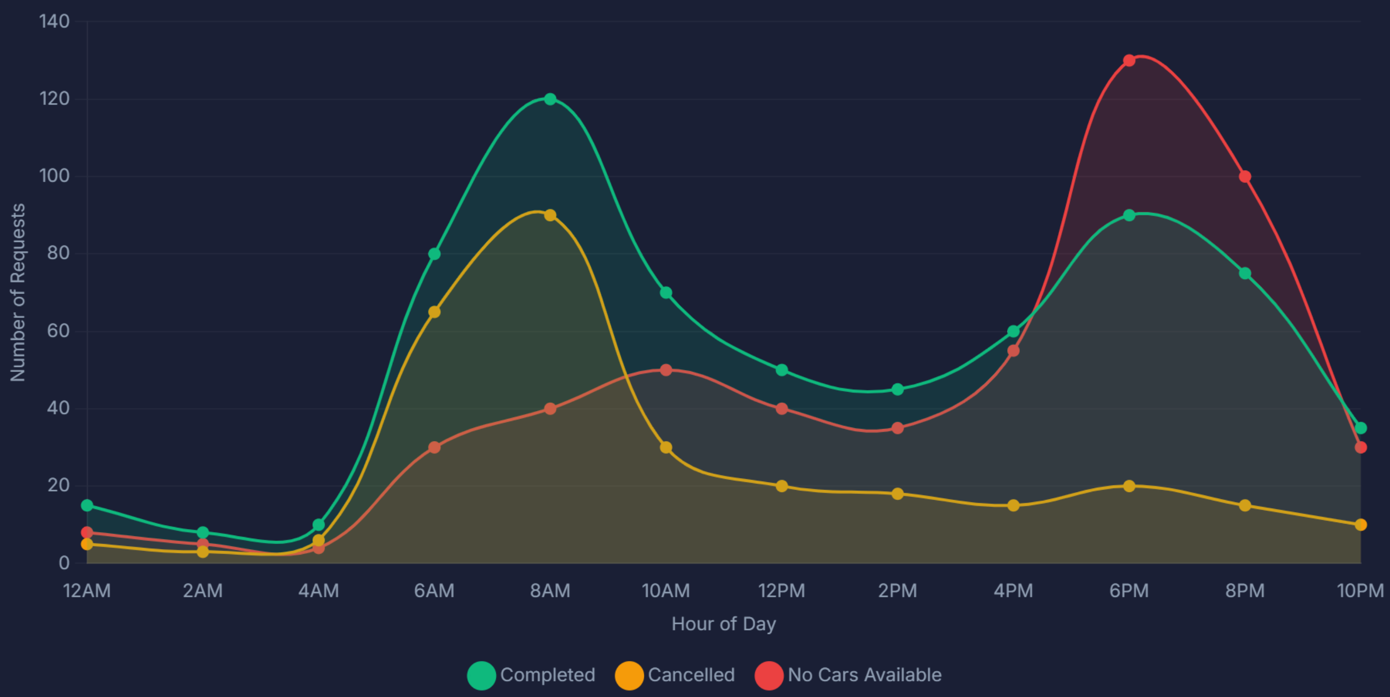
Identifying peak customer demand hours throughout the day



💡 **Key Insight:** Morning (5-9 AM) and Evening (5-9 PM) office commute hours had the highest ride demand.

Chart 5 — Ride Status Breakdown by Hour

Supply-demand imbalance throughout the day



💡 **Key Insight:** Cancellations peaked during morning hours. "No Cars Available" peaked during evening hours — revealing a dual supply gap.

Chart 6 — Ride Requests by Time & Pickup Point

Demand patterns across locations and time periods

💡 **Key Insight:** Airport demand surged during Evening/Night periods. City demand peaked during Morning office hours.

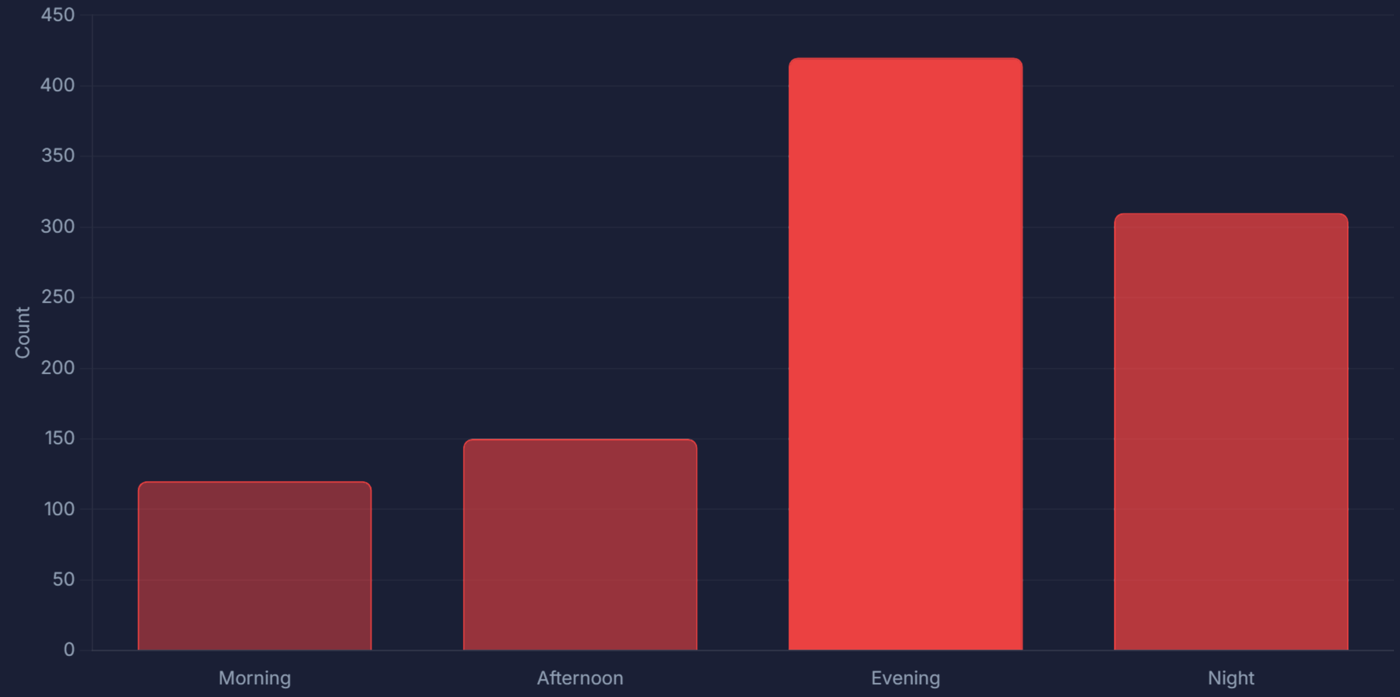
Chart 7 — Cancelled Rides by Pickup Point

Cancellation trends across pickup locations

💡 **Key Insight:** City pickup points recorded significantly more ride cancellations than Airport locations.

Chart 8 — "No Cars Available" Cases by Time Slot

Identifying peak cab shortage periods



💡 **Key Insight:** Evening hours experienced severe driver shortages — the most critical operational gap identified.

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Major Business Findings

Critical insights derived from the analysis

1. Uber faces a **major supply-demand imbalance** during peak office commute timings.
2. Evening hours experience the **highest "No Cars Available"** cases.
3. Morning hours experience **higher ride cancellations** from drivers.
4. Airport pickup locations are heavily affected by **driver shortages**.
5. Ride demand strongly follows **daily office commute patterns**.
6. Driver availability **does not scale** efficiently during peak demand.

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Business Recommendations

Actionable strategies to address the supply-demand gap

1

Increase Evening Driver Coverage

Deploy more drivers during Evening and Night hours to address the critical "No Cars Available" gap during peak demand.

2

Driver Incentive Programs

Offer Morning peak-hour incentives and bonuses to reduce driver-side cancellations and improve ride acceptance rates.

3

Dynamic Pricing Strategy

Implement surge pricing during high-demand periods to balance supply-demand and incentivize driver availability.

4

Location-Based Driver Allocation

Improve driver distribution between Airport and City zones using real-time demand prediction algorithms.

5

Shift-Based Scheduling

Introduce structured driver shifts aligned with Morning and Evening commute hours to ensure consistent coverage.

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Conclusion

Summary of the end-to-end analytics workflow and outcomes

This project successfully identified major operational inefficiencies within Uber's ride request system. The analysis showed that:

- Customer demand **exceeds driver availability** during Morning and Evening peak hours
- Driver shortages and ride cancellations **significantly affect customer experience**
- Supply-demand balancing strategies can **improve ride completion rates** and operational efficiency

🔄 End-to-End Workflow

📑 Data Cleaning → 🗄️ SQL Analysis → 📊 EDA → 📈 Visualization → 💡 Recommendations

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Skills Demonstrated

Core competencies applied throughout this project

🧹 Data Cleaning | 🗄️ SQL Querying | 📊 Exploratory Data Analysis | 📈 Data Visualization | 💼 Business Problem Solving | 🧠 Analytical Thinking | 🔍 Data Interpretation | 📝 Reporting & Documentation